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Author(s): Edward F. Neuhauser, Roy Hartenstein and David L. Kaplan

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Growth of the earthworm *Eisenia foetida* in relation to population density and food rationing

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At 25°C *Eisenia foetida* hatches from its cocoon about 3 weeks after fertilization and follows a logistic growth pattern. It grows slowly for about three weeks and then enters a rapid phase whose slope is useful as an index to the nutritional quality of its food. The rapid phase is followed by a steady state maintenance phase, or by a phase of weight decline whose slope can be used as an index to a property of the food which is related to starvation latency. The slope of the rapid phase for individually grown worms was steeper on activated sludge than on cow manure and steeper on the latter than on horse manure. The maximum weight achieved by individually tested *E. foetida* was significantly greater with activated sludge than with either manure as food. The manures provided nutrients at a rate suboptimal to genetic capability for growth but either served as a better source of food than sludge when submaintenance levels remained, or were less toxic as castings. In a volume of about 30 cc over an area 78 cm² the carrying capacity of 250 g horse manure on 50 g silt loam was 6 g live weight *E. foetida*; this value for these periodically disturbed systems is considerably less than what can be achieved in undisturbed systems published elsewhere. Carrying capacity was not achieved in the same volume-area-time relation with activated sludge in place of manure, though more than 23 g could be supported; more than ten times as much biomass of *E. foetida* can be supported per unit area of soil overlain by activated sludge than has been reported for earthworms in natural ecosystems.

E. F. Neuhauser, R. Hartenstein and D. L. Kaplan, School of Biology, Chemistry and Ecology, State Univ. of New York, College of Environmental Science and Forestry, Syracuse, NY 13210, USA.

При 25°C *Eisenia foetida* выходит из кокона примерно через 3 недели после оплодотворения и имеет логистическую кривую роста. Она растет медленно примерно в течение 3-х недель, а затем наступает быстрая фаза, в которой наклон кривой используется в качестве индекса для оценки питательных свойств корма. После быстрой фазы наступает стабилизированная фаза или фаза потери веса, на которой наклон кривой используется как индекс для оценки свойств пищи, определяющих латентное голодание. Наклон кривой в быстрой фазе у червей при индивидуальном содержании круче на активированном отстое, чем на коровьем навозе, а на последнем – круче, чем на конском навозе. Максимальный вес, достигаемый *E. foetida* при индивидуальном содержании в активированном отстое выше, чем в навозе. Навоз содержит элементы питания в количестве, субоптимальном для реализации генетической способности к росту, но, либо является лучшим видом пищи, чем отстой, для достижения уровня, близкого к максимальному, либо менее токсичен, чем отстой. В объеме около 20 см³ на площади 78 см² 250 г конского навоза, смешанного с 50 г илистой глины, дают 6 г живого веса *A. foetida*. Данная величина для этих, периодически разрушающихся систем меньше, чем та, которая может достигаться в ненарушаемых системах, описанных повсеместно. Поддерживающая способность не достигалась в том же соотношении объема – площади – времени при замене навоза отстоем, но более чем 23 г могут быть обеспечены; в 10 раз большая биомасса *A. foetida* может быть достигнута на единицу площади почвы, покрытой активированным илом, чем та, которая была установлена для червей в естественных экосистемах.

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1. Introduction

Serious problems may be encountered in the disposal of cattle manures (Magdoff et al. 1978) and sewage sludges (Information Transfer 1977). Since a potential exists for using the earthworm *Eisenia foetida* (Savigny) in managing these wastes (Graff 1974, Watanabe and Tsukamoto 1976, Hartenstein et al. 1979) and obtaining a by-product rich in protein (Schulz and Graff 1977, Sabine 1978), a better understanding of its biology and ecology is needed. Data are presented in this paper on the growth of *E. foetida* on horse manure, cow manure, and activated sludge. Emphasis is placed on the growth curve of individual specimens provided with an abundance or ration of food, and populations of *E. foetida* in a limited volume of food.

2. Materials and method

Young *E. foetida* (6–10 mg) were obtained from a stock culture in our laboratory and grown at 25°C on the food sources indicated. Activated sludge was obtained with about 1% solids from an aeration tank at the Meadowbrook-Limestone Wastewater Treatment Plant, Onondaga County, NY. The sludge was concentrated to about 11% solids by centrifuging in a Sorvall SS3 with a GSA rotor at 5000 rpm for 10 min. Horse and cow manure were obtained fresh as needed, uncontaminated by urine. Growth rates for individual worms at 25°C with an unlimited supply of food were determined by placing them separately into 20 × 100 mm petri dishes with a 5 cm layer of soil as substrate with food on top and weighing on days indicated in results. Growth rates of individuals on food rations were determined similarly, except for the limitation of food. The volume-area of this system was 100 cc – 78 cm². Growth rates of populations of worms were determined by placing them into 10 cm diameter × 5 cm high dishes containing 50 g of soil and 250 g manure or activated sludge. The volume-area of this system was 300 cc – 78 cm². Numbers of replicates for each experiment are given in figure captions. Weights are based on survivors only. The worms were weighed without first voiding them, since it was known that the gut content would lie around 10% of live weight, whereas larger differences were expected in relation to treatment, and stress may attend attempts to void the worms. Corrections for gut content were not applied to any of the data in this study.

3. Results

3.1. Mortality

Survival was a mean of 85% at the conclusion of studies indicated for setups with individually cultured worms (Fig. 1) and 91% after six months with populations (Figs 2, 3 and 6). Mortality which occurred could not

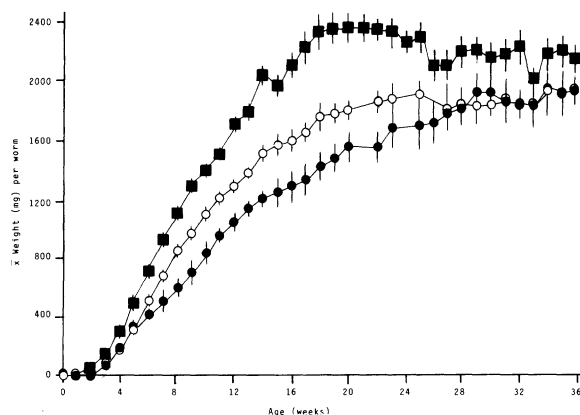


Fig. 1. Growth of individual *E. foetida* in an area 78 cm² with 50 g soil overlain with 250 g cow manure (○), horse manure (●) or activated sludge with 11% solids (■). Food source was replaced weekly. N = 7 for cow manure, 15 for horse manure, and 20 for sludge. Bars are SEs.

necessarily be attributed to handling, substrate, or population density.

3.2. Growth of individuals with unlimited food

At 25°C *E. foetida* emerges from its cocoon three weeks after fertilization and exhibits a growth pattern which follows the logistic curve of Verhulst. Independently of substrate, newborn added weight slowly for 3 wk and then rapidly and linearly with time for at least 5 wk (Fig. 1). It is understood, of course, that this linearity is only apparent, not real. Here we take it as an approximation of the slope of the logistic. After 8 wk of age a clear departure from apparent linearity was observed for specimens on cow manure, a week thereafter for *E. foetida* on activated sludge, and an additional 3 wk later for individuals on horse manure. The slopes of the linear, or rapid, phase of absolute growth, based on linear regressions over the age period 3 to 8 wk on cow manure, horse manure, and activated sludge, 164, 103, and 204 mg wk⁻¹, were significantly different ($p < 0.01$). At age 10 wk the mean $\pm$ SE weight of worms grown in these substrates was 1132 ± 43 , 871 ± 75 , and 1454 ± 47 mg respectively; differences between these means also were significant ($p < 0.01$). Significance was not obtained for the maximum weight achieved ultimately, about 1.9 g upon horse manure or cow manure as food, though this weight was significantly less than what was ultimately achieved on activated sludge, about 2.4 g. Clearly, the activated sludge was superior to either manure as food for *E. foetida*.

3.3. Growth in relation to population density

Figs 2 and 3 show relations between growth of *E. foetida* and population density with horse manure and activated sludge as food. Again it is seen that the acti-

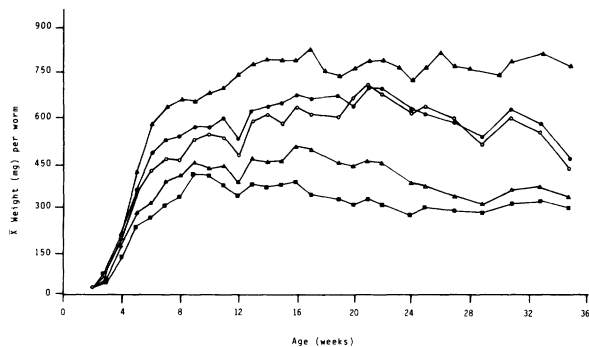


Fig. 2. Growth of populations of *E. foetida* in an area 78 cm² with 50 g soil overlain with 250 g horse manure. Population density per dish (N = 3) from uppermost to lowest curves were 3, 4, 8, 12, and 16 worms per 300 cc. Curves for densities of 5 and 7 were not included for the sake of clarity.

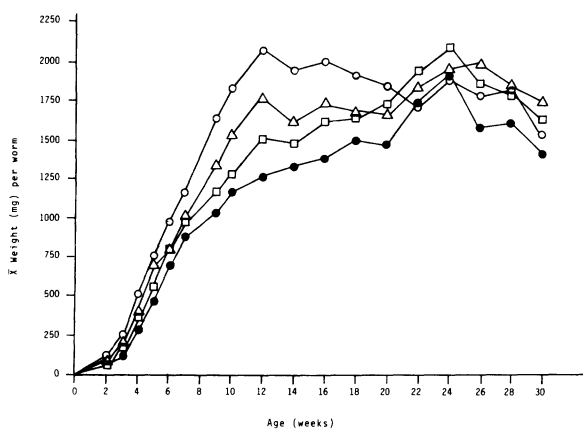


Fig. 3. Growth of populations of *E. foetida* in an area 78 cm² with 50 g soil overlain with 250 g activated sludge with 11% solids. Population density per dish (N = 3) from uppermost to lowest curves were 4, 8, 12 and 16 worms per 300 cc.

activated sludge was a superior food source. For insight into the effect of population density on growth, slopes of the rapid growth phase were calculated from the mean weights at ages 3 and 5 wk, during which the rate of absolute growth was linear with time. The y ordinates for one worm in Fig. 4 show that the theoretical maximum growth rates in horse manure and sludge, based on unlimited space and food, are 160 and 288 mg per worm per week, respectively. When four or more worms were present per 300 cc on an area of 78 cm², these rates were reduced by a factor of 7.81 and 4.68 per worm in accord with the linear regressions: $R_{\text{sludge}} = -7.81D + 296$, $r = -0.93$, and $R_{\text{manure}} = -4.68D + 165$, $r = -0.93$, where R is mg worm⁻¹ wk⁻¹ and D , density (Fig. 4).

For insight into the effect of population density on production, the mean of the maintenance weights achieved between ages 14 and 17 wk with horse manure and ages 14 and 18 wk with activated sludge were mul-

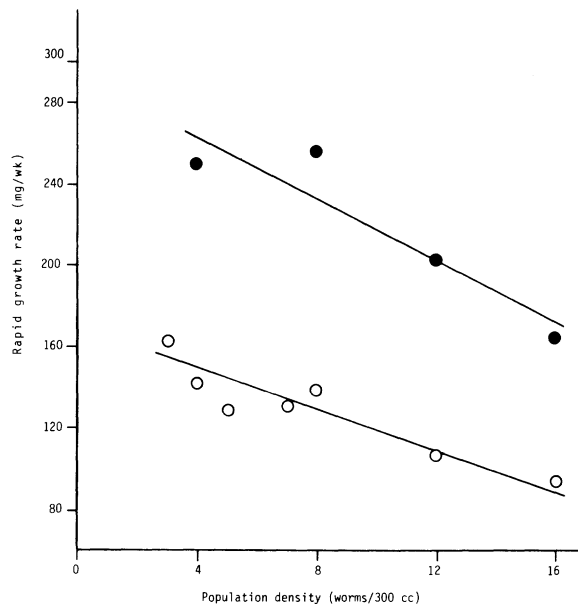


Fig. 4. Slopes of rapid growth phase of *E. foetida* in relation to population density in an area 78 cm² with 50 g soil overlain with 250 g horse manure (○) or activated sludge with 11% solids (●). Data derived from Figs 2 and 3.

tiplied by density and plotted as biomass versus population density (Fig. 5).

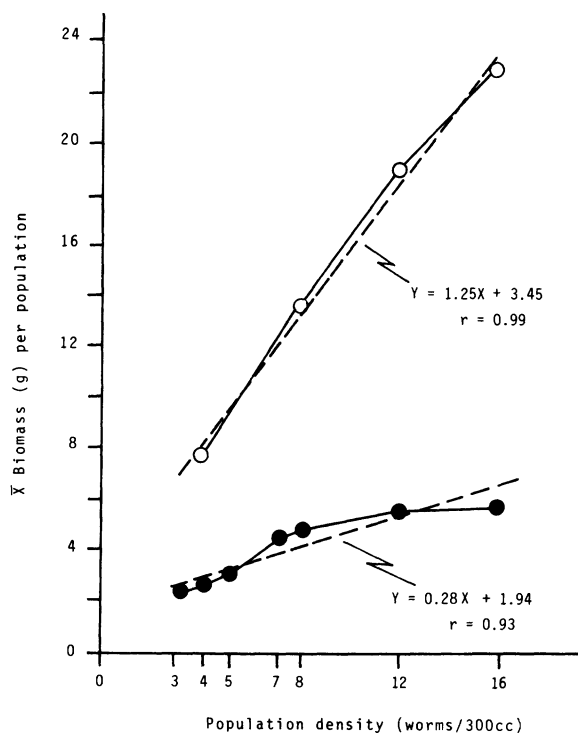


Fig. 5. Maximum biomass of *E. foetida* in relation to population density in an area 78 cm² with 50 g soil overlain with 250 g horse manure (●) or activated sludge with 11% solids (○). Data derived from Figs 2 and 3.

Tab. 1. Linear regressions of weight in mg wk⁻¹ (W) on age in weeks (A) for growth of *E. foetida* on limited amounts of horse manure, cow manure or activated sludge at 25°C.

Food	dry wt (g)	Age 2–8 wk	r	Age 9–16 wk	r
Horse manure	1.10	W = 47.9A – 20.5	0.96	W = –10.5A + 430.3	–0.90
	2.20	W = 61.4A – 47.4	0.97	W = – 6.3A + 537.4	–0.66
	3.30	W = 93.8A – 105.5	0.99	W = – 9.1A + 737.9	–0.68
	4.40	W = 95.9A – 112.6	0.99	W = –12.6A + 828.7	–0.69
Cow manure	1.05	W = 33.6A – 44.4	0.89	W = –12.6A + 381.2	–0.95
	2.10	W = 58.1A – 45.2	0.95	W = – 16.8A + 551.8	–0.91
	3.15	W = 96.6A – 114.3	0.98	W = –17.5A + 797.0	–0.98
	4.20	W = 109.2A – 164.3	0.99	W = – 9.8A + 789.0	–0.70
Sludge	0.50	W = 28.0A – 35.4	0.90	W = –14.0A + 348.4	–0.99
	1.00	W = 51.8A – 39.7	0.96	W = –14.0A + 452.1	–0.99
	1.50	W = 83.3A – 114.4	0.99	W = –14.0A + 600.0	–0.99
	2.00	W = 100.8A – 167.8	0.98	W = –14.7A + 714.5	–0.99

The biomass achieved in horse manure had reached a limiting value of about 6 g live weight *E. foetida*, while the biomass achieved in activated sludge had not approached a limiting value (Fig. 5); the data show, however, that at least 23 g of *E. foetida* could be obtained when activated sludge is used in place of manure in the same space-time dimension.

3.4. Growth in relation to availability of food

Graphs of growth versus time on various rations of activated sludge are provided in Fig. 6. The ascending and descending components of these curves, and for similar curves obtained with both cow and horse manures, can be expressed as linear regressions (Tab. 1). Plots of the slopes of the rapid growth phases of the ascending components for ages 3 to 5 wk are given in Fig. 7 along with regression equations which relate the slopes to ration. The slopes achieved with horse manure were lower than

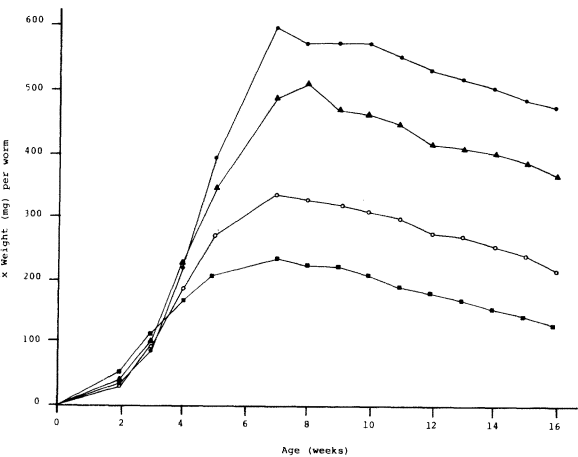


Fig. 6. Growth of *E. foetida* from cocoon emergence to 16 weeks on 5 g (●), 10 g (▲), 15 g (○) or 20 g (■) of activated sludge (11% solids). N = 3.

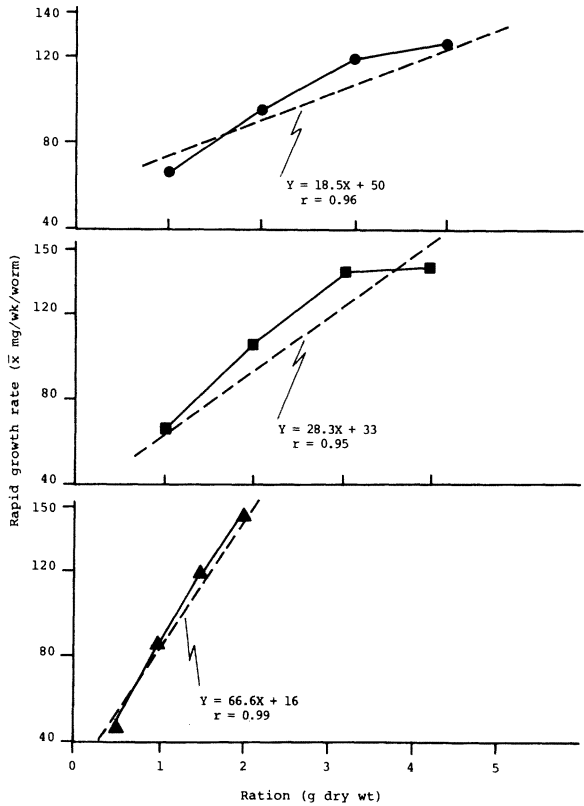


Fig. 7. Rapid growth rate of *E. foetida* in relation to ration of horse manure (●), cow manure (■) or activated sludge initially with 11% solids (▲).

with cow manure which in turn were lower than with activated sludge. Expressed as a ratio, 1:1.5:3.6, it can be seen that the worms grew 3.6 times more rapidly over the range of 0.5 to 2.0 g (dry wt) of activated sludge than in 1.1 to 4.4 g (dry wt) of horse manure.

From the equations of the descending limbs of the curves (Tab. 1), calculations were made of percent loss in live weight per day from age 63 to 112 d. Plots of the resulting values versus ration are given in Fig. 8. These

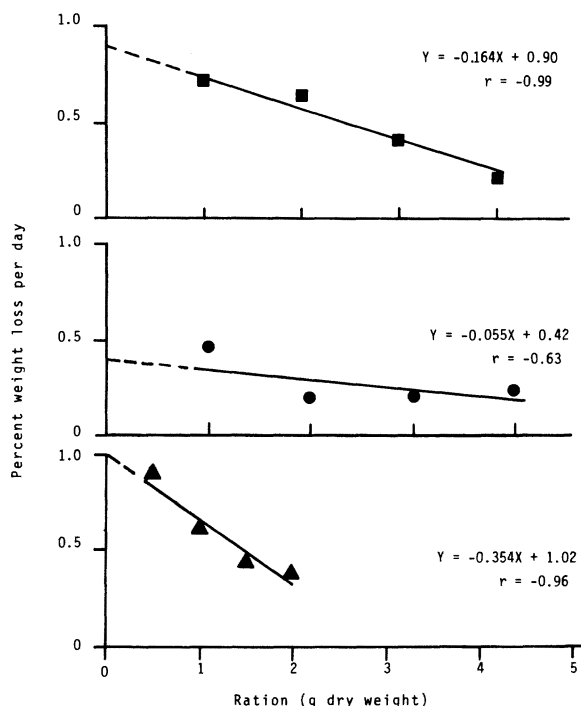


Fig. 8. Percent loss in live weight of *E. foetida* per day in relation to ration of cow manure (■), horse manure (●) or activated sludge (initially with 11% solids) (▲) in the age interval of days 63 to 112 after emergence from a cocoon.

values may be compared with a weight loss of 1.3% per day at 25°C for 37 d obtained in a separate study in which *E. foetida* was fasted upon ashed loam (unpubl.). Percent weight loss in relation to initial weight of ration occurred more rapidly for worms that were fed on sludge: % W_{loss} sludge = $-0.35D + 1.02$, $r = -0.96$, than on cow manure, % W_{loss} cow manure = $-0.1D + 904$, $r = 0.99$, where D is population density. Percent loss in weight was least for worms that had fed upon horse manure, % W_{loss} horse manure = $-0.055D + 0.42$, $r = -0.63$, and appeared independent of initial ration of horse manure beyond the lowest level offered.

4. Discussion

4.1. Growth rates of *E. foetida* and other earthworms

Previous experiments on growth of *E. foetida* at 25°C in cow manure (Graff 1974) or "field compost" (Tsukamoto and Watanabe 1977) from cocoon emergence to age 10 wk resulted in specimens weighing 600 to 800 mg. These values lie below what was obtained in this study for individually grown worms on unlimited food, but fall within the range observed in the population study with horse manure as food. Clearly, in the presence of conspecifics *E. foetida*'s growth rate is severely retarded. This is due not only to competition for food, and because a large proportion of energy for

tissue production by worms grown in isolation is shunted into production of cocoons by worms grown in populations (Hartenstein et al. 1980) but also because the egesta of *E. foetida* is extremely toxic to the species (Kaplan et al. 1980).

Populations of *E. foetida* gain weight at rates which are dependent on population density. Approximately 300 to 800 mg of biomass were produced per individual from birth to about 6 wk age in horse manure at 25°C in this study. The corresponding growth rates, 50 to 133 mg wk⁻¹, may be compared to growth of other earthworms. Michon (1954) reported a rate of about 90 mg wk⁻¹ for 10 wk at 18°C in culture for *Dendrobaena subrubicunda*, and graphs given by Lakhani and Satchell (1970) show production of about 130 mg wk⁻¹ during periods of rapid activity for *Lumbricus terrestris* under varying temperatures around 13°C. Only about 3.5 mg wk⁻¹ is produced by *Allolobophora rosea* however, "under near normal field conditions" (Phillipson and Bolton 1977).

4.2. Growth in relation to food type and population density

The upper limits to the rapid growth phase (Fig. 4) show that *E. foetida* grows about 1.7 times more rapidly in sludge than in horse manure. The hypothetical maximum for one worm in sludge, 288 mg wk⁻¹, is 40% greater than the value 204 mg wk⁻¹ obtained in the experiments with single worms given an unlimited supply of food and space (Fig. 1). The maximum hypothetical value for horse manure, 160 mg wk⁻¹, also exceeds the value, 103 mg wk⁻¹, obtained experimentally for single worms (Fig. 1). As the population density of *E. foetida* was increased, an inhibitory effect on growth rate was observed. This effect was more pronounced in horse manure than in sludge, as indicated by the 2.9 fold greater slope of the rapid growth phase in sludge at the highest population density in contrast to the corresponding value of 1.7 denoted above (Fig. 4).

In addition to population density as a factor limiting the rate of growth during the rapid growth phase, the nature of manure as a substrate, in contrast to sludge, appears to be a limiting factor. This is seen in Fig. 7, where the rate of the rapid growth phase increases linearly with an increase in ration of activated sludge and appears to be tailing off with an increase in ration of either cow or horse manure. These differences in response may be attributed to the higher concentration of microbes per unit volume of sludge (Curds and Hawkes 1975) relative to manures (Hrubant et al. 1978).

Differences between quality of food with respect to tissue production are also seen in Fig. 7, where comparisons of maximum growth rates linear with time are shown in relation to food rationing. Since *E. foetida* transmits food from mouth to periproct in about three hours independently of nature of ingesta (unpubl.), its growth rate is governed, in part, by the nutritional quality of the quantum of food transmitted. The slopes of

the plots in Fig. 7 show that *E. foetida* grew 3.6 times faster during ages 3 to 5 wk on activated sludge than on horse manure. The activated sludge used was obviously nutritionally superior to the horse manure.

At the highest ration of horse manure (4.4 g dry wt), the steepest slope of the rapid growth phase, 125 mg wk⁻¹, exceeded the value obtained in Fig. 1 under conditions of unlimited space and food, though this rate of increase lasted only 2 wk from age 3 to 5 wk. With cow manure, the highest ration, 4.2 g, allowed a maximum value of only 142 mg wk⁻¹ between ages 3 and 5 wk, versus 164 mg wk⁻¹ shown in Fig. 1. Similarly, the maximum rate of growth during the rapid growth phase in sludge, 145 mg wk⁻¹, was less than what was achieved under the less restricting conditions related to Fig. 1.

When *E. foetida* receives food below a maintenance level, it loses weight at a rate which appears to depend upon the quantity and nature of its ingestible substrate. In the total absence of organic matter, weight loss proceeds at a rate of 1.3% of body weight per day for 37 d at 25°C (unpubl.). In dishes initially supplied with 0.50, 1.05 and 1.10 g (dry wt) of sludge, cow manure and horse manure, respectively, losses of 0.9, 0.7 and 0.5% per day were encountered after maximum weight was achieved and most of the food had been consumed. Lower rates of loss were exhibited in dishes with higher initial quantities of these foodstuffs (Fig. 8). The lowest losses were found among worms in the manures, in contrast to sludge. The manures thus provided nutrients to the worms at a rate suboptimal to worm growth in sludge, but either served as a better source of food than sludge when less than a maintenance level of either material remained, or were less toxic as castings. The latter explanation is related to the fact that earthworms do not practice coprophagy; their castings are toxic to themselves and other species (Kaplan et al. 1980).

4.3. Carrying capacity

From Fig. 5 it is possible to deduce a carrying capacity for horse manure, a value of about 6 g *E. foetida* per 300 cc of manure and soil on an area of 78 cm². This value should not be construed as the true carrying capacity of this system; a substantially higher value, 9.6 g/110 cc on an area 24 cm², was found in another study (unpubl.), in which the culture medium of *E. foetida* was left undisturbed for a period of 7 wk, and the experiment designed allowed a valid estimate of carrying capacity. Fig. 5 also shows that carrying capacity was not achieved in the activated sludge, but that its value exceeds 23 g/78 cm². The activated sludge was thus capable of supporting about four times more biomass of *E. foetida* than could be carried by the same volume of horse manure. Calculations of linear regression equations for these data (Fig. 5) show that *E. foetida* can produce its biomass about 4.4 times more rapidly in activated sludge than in horse manure.

Estimates of earthworm biomass in fields indicate

maximum amounts of about 280 g m⁻² (Edwards and Loftly 1977). Data from Fig. 5 suggest that more than tenfold as much biomass, 2900 g of *E. foetida*, could be supported in one square meter if sufficient activated sludge is present.

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